Correct Verdicts, Wrong Field: Attribution and Capability Are Dissociable When Small Language Models Follow a Rubric

Sophie D. Neilson
Independent Researcher
ORCID: 0009-0001-2430-1743

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Abstract

A behavioral format assay changes the structure of a prompt and reads whether a model's answers improve. Verdict accuracy is the usual measure. We show that it is not sufficient. We gave three local language models a structured decision rubric that names an explicit decision field, and asked each for a verdict and for the field that determined it. Across six decision classes and two scenario polarities, verdict accuracy sat at or near ceiling (0.96–1.00), while the share of correct verdicts that cited the intended decision field ranged from 0.05 to 0.66. On the firing scenarios, only 27% of correct verdicts cited the intended field; the rest routed through a recognition axis the rubric does not designate as the decision route. One 32B model answered correctly in every run while naming the intended field in five of ninety-six. A grader scoring only verdicts would credit the rubric's decision field for a result the model attributes elsewhere. We then removed the two recognition axes the models were routing through. Accuracy did not fall. The models re-routed to the intended field and stayed correct: citation of the intended field rose from 63 to 96 runs for the thinking model and from 5 to 65 for the 32B model. The shortcut is a preference, not a crutch. The intended path was available throughout. Attribution is dissociable from capability: the field a model reports is not the only field it can use, and neither the verdict nor the self-report is a reliable proxy for the other. We argue that a format assay must track the reasoning route, not only the verdict, and that a model's self-reported route must itself be checked against an intervention rather than trusted.

1 Introduction

A common way to test whether a change to a prompt's structure helps a language model is to score the model's answers before and after. If accuracy rises, the structure gets the credit. This paper is about a failure of that inference. A model can reach the correct answer through a part of the prompt that the change was not about, so that a rise in accuracy is not evidence that the intended structure did any work.

We study this with a structured decision rubric. The rubric describes a class of decision an agent faces, and it names one field, a scope condition, as the field on which the verdict turns. It also carries two recognition axes that describe what a violation and a correct action look like. The rubric states the order of evaluation: read the scope condition, then render the verdict. We ask a model for a verdict on a concrete scenario and, on the same turn, for the name of the field that determined its verdict. The verdict is scored against ground truth. The named field is scored against the field the rubric designates.

Two questions follow. First, does verdict accuracy track engagement with the intended field? We find it does not: accuracy is near ceiling while intended-field citation is far lower, and the gap is largest on the scenarios where the decision class fires. Second, is a correct-but-off-route answer fragile: does it depend on the field the model cites? We test this by removing the recognition axes the models route through. It is not fragile. Accuracy holds and the models re-route to the intended field. The route a model names is a preference, not a requirement.

We call the resulting distinction *attribution versus capability*. What a model reports as the basis of its answer (attribution) is separable both from whether the answer is correct (verdict accuracy) and from which fields the model can actually use to be correct (capability). A behavioral format assay that reads only the verdict sees none of this. It cannot tell a genuine effect of the intended structure from a rise carried by an unintended field, and it cannot tell a load-bearing field from one the model merely names.

1.1 Related Work

That a model's stated reasoning need not reflect the cause of its answer is established for chain-of-thought prompting. Turpin et al. [2023] show that models give plausible reasoning that does not mention the features actually driving their predictions, and Lanham et al. [2023] measure how far a stated chain of thought is load-bearing for the final answer. Chen et al. [2025] report the same for reasoning models, whose stated chains omit decisive influences such as an injected hint. Our field-source measure is a compact form of the same question for a rubric-following task: the model names the field it used, and we check that name against an intervention rather than taking it at face value. The distinction between an answer's correctness and the process that produced it also motivates process-level supervision, where a reward is placed on the reasoning steps rather than the outcome [Lightman et al., 2023]. Our result is a reason that outcome scoring alone is insufficient for a format study, framed as a measurement problem rather than a training one.

A model reaching the right answer through an unintended feature is the shortcut-learning pattern [Geirhos et al., 2020], described there for learned representations; we observe it at the level of a model's self-reported field on a structured prompt, and we add that the shortcut here is not load-bearing: the intended route survives its removal. The rubric we use comes from a research program on descriptive decision-point prompts [Neilson, in preparation]; two adjacent studies of format-dependent behavioral shifts in language models [Wang et al., 2026, Liu et al., 2026] vary prompt structure and read behavioral output, but neither separates the model's self-reported route from its verdict, which is the measure this paper turns on.

2 Materials and Method

2.1 The Rubric and the Field-Source Measure

Each decision class is described by a rubric with a fixed shape. A **Y − Decision terrain** block names the evaluation order and three parts: a **Pull character** (why the locally-wrong action is attractive, present in both firing and non-firing cases), a **Scope-operative-when** field (the condition under which the decision class fires, with a verification procedure), and a **Scope-not-operative-when** field (the condition under which it does not, stated as a positive discriminator). Below the terrain sit two axes: a **Marker axis** (an invariant describing what the violation looks like) and an **Aim axis** (an invariant describing correct navigation). A **Rest axis** describes territory where the class does not arise. The terrain block states: "Evaluate in sequence: pull character → scope → verdict. Do not render a verdict before completing the scope check." The Scope field is therefore the field the rubric designates as the decision route; the Marker and Aim axes are recognition and characterization, not the route.

We use six decision classes (Section B). Each has two scenarios: a *firing* scenario whose ground truth is "yes" (the class applies) and a *carve-out* scenario whose ground truth is "no" (a named exception holds). The intended field for a firing scenario is Scope-operative-when; for a carve-out scenario it is Scope-not-operative-when. Both are the Scope field.

The instrument is the response format. The model is asked to answer in four lines: a `VERDICT` (yes/no), a `FIELD SOURCE` (the name of the field that determined the verdict), an `OBSERVABLE` (the condition it cites), and the `EVIDENCE` (the trace element it rests on). The verdict and the field source are parsed deterministically. A *wrong-path success* is a correct verdict whose cited field is not the Scope field.

2.2 Model and Sampling

Three subjects, all Q4_K_M GGUF quantizations served by llama.cpp (`llama-server`, build `b9445-af6528e6d`) over the raw completion endpoint, so no server-side chat template sits between the declared prompt and the model. Each model's instruct wrapper is published verbatim (Appendix A).

- **Mistral-7B-Instruct-v0.3** (7.2B), non-thinking.
- **Qwen2.5-32B-Instruct** (32B), non-thinking.
- **Qwen3.8-27B** (27B), thinking enabled. Its reasoning trace is recorded per run.

Claude-family models are excluded as subjects: the rubric corpus is in their training data. The sampler chain is declared complete and identical across subjects: temperature 0.8, `min_p` 0.05 as the sole truncation stage, every other truncation stage and every penalty disabled, per-run seeds 1–8. Penalties are off by design: the rubric induces repeated structure (axis-by-axis traversal), and a repetition penalty would attenuate the behavior under study. The served context was 32768 tokens for Mistral and Qwen3.8. Qwen2.5-32B was served at 8192 tokens: its weights plus the KV cache at 32768 exceed the 24 GB GPU. The prompts are about 3800 tokens with at most 1024 generated, so 8192 has ample headroom and no run truncated; the served context is recorded per run and bounds only the unused tail.

2.3 Conditions and Arms

Every run assembles one prompt: the shared rubric-definition document, then the decision-class terrain, then the scenario trace, then the four-line instruction. The study has two arms that differ only in the terrain:

- **Base arm.** The full terrain, with the Marker and Aim axes present.
- **Ablation arm.** The same terrain with the Marker and Aim axes removed. The Y-Decision block (including both Scope fields), the Pull character, and the Rest axis are byte-identical to the base terrain; a per-file diff confirms only the two axes changed.

The design is 3 subjects $\times$ 2 arms $\times$ 6 classes $\times$ 2 scenarios $\times$ 8 runs $=$ 576 runs. No run truncated, and the served model matched the declared model in every run.

2.4 Scoring

The headline measures are deterministic. The verdict is parsed and compared to ground truth. The field source is parsed and classified by the first field-family keyword it names: Scope, Marker, Aim, Pull, or other. For each cell (subject, arm, class, scenario) we report verdict accuracy over the eight runs, and among the correct verdicts the share that cited the Scope field. A run's answer is the four-line response; for the thinking model the reasoning trace precedes it and is recorded separately, so the four parsed lines are the same shape for all subjects. Read cells and direction, not exact counts: at eight runs a single-integer difference is inside the noise band.

2.5 Agents and Models

Table 1 lists every model and agent with a role in producing the results. The three subjects are the models under test. A Claude session authored the study and the analysis code; it is an agent, not a person, and it judged nothing in the deterministic layer, which is a parser.

Table 1: Agents and models.

Role	Model	Version and runtime	Date
Subject (non-thinking)	Mistral-7B-Instruct-v0.3	Q4_K_M, llama.cpp b9445, ctx 32768	2026-09-11
Subject (non-thinking)	Qwen2.5-32B-Instruct	Q4_K_M, llama.cpp b9445, ctx 8192	2026-09-11
Subject (thinking)	Qwen3.8-27B	Q4_K_M, llama.cpp b9445, ctx 32768, thinking on	2026-09-11
Study and analysis author	Claude (agent)	Version not recorded	2026-09-11

3 Results

3.1 Verdict Accuracy Overstates Field Engagement

Verdict accuracy is at or near ceiling for every subject, while the share of correct verdicts that cite the intended Scope field is far lower (Table 2). The models are right, and they largely say a different field made them right.

Table 2: Base arm, per subject, over 96 runs each (12 cells $\times$ 8).

Subject	Verdict accuracy	Scope-cited / correct	Wrong-path / correct
Mistral-7B	0.96 (92/96)	0.66	0.34
Qwen2.5-32B	1.00 (96/96)	0.05	0.95
Qwen3.8	1.00 (96/96)	0.66	0.34

Qwen2.5-32B is the sharpest case. It answered correctly in all 96 runs and named the Marker axis in 88 of them, the Scope field in five. A grader scoring only its verdicts would read a perfect score and credit the scope structure for a result the model attributes almost entirely to a recognition axis.

The gap is widest on the firing scenarios. Pooling the base arm across subjects, verdict accuracy is 0.99 on both polarities (142/144 each), but intended-field citation among correct verdicts is 0.27 on the firing scenarios against 0.64 on the carve-outs. When the class applies, roughly three of four correct "yes" verdicts are reached by naming the Marker axis's firing condition rather than the scope condition the rubric puts on the decision path.

3.2 Removing the Shortcut Does Not Break the Answer

If a correct verdict depended on the recognition axis the model cited, removing that axis should collapse the cell. It does not. In the ablation arm, with the Marker and Aim axes deleted, verdict accuracy holds at ceiling for all three subjects. The base-versus-ablation accuracy differential between shortcut-routing cells and scope-routing cells (classified from the base arm) is within noise of zero for every subject; the only movements are two small *improvements* under ablation for Mistral (one carve-out cell 6/8 $\rightarrow$ 8/8, one firing cell 6/8 $\rightarrow$ 7/8).

What changes is the cited field. With the recognition axes gone, the models route to the Scope field and stay correct (Table 3). The thinking model cites the Scope field in all 96 ablation runs, up from 63; the 32B model rises from 5 to 65.

Table 3: Field-source class over all 96 runs per subject per arm. Verdict accuracy is unchanged between arms (near ceiling throughout).

Subject	Base arm	Ablation arm
Mistral-7B	scope 63, pull 15, marker 9, aim 1, other 8	scope 73, pull 21, other 2
Qwen2.5-32B	marker 88, scope 5, other 3	scope 65, other 31
Qwen3.8	marker 32, scope 63, other 1	scope 96

The intended path was available throughout. The models preferred the recognition axis when it was present and used the scope condition when it was not, at no cost to accuracy. The shortcut is a preference, not a crutch.

4 Discussion

4.1 Attribution Is Not Capability

Three quantities that a verdict-only assay silently conflates come apart here. The *verdict* is whether the answer is correct. The *attribution* is the field the model names as its basis. The *capability* is which fields the model can in fact use to be correct. Verdict accuracy is near ceiling regardless of the other two. Attribution is mostly off the intended field in the base arm. Capability includes the intended field the whole time, as the ablation shows. Knowing the verdict tells you nothing about the attribution, and knowing the attribution tells you nothing about the capability.

For a behavioral format assay this is a direct constraint. A rise in verdict accuracy after a structural change is not evidence that the intended structure did the work: the model may reach the answer through another field, as ours did three times in four on the firing scenarios. And a field the model never names may still be one it can use, so the absence of a citation is not evidence of an absent capability. The assay must track the route, and it must not read the model's self-reported route as ground truth.

4.2 The Self-Report Must Be Checked, Not Trusted

The field-source line is the model's account of its own decision. Taken at face value it would say that Qwen2.5-32B is a Marker-axis reasoner that essentially ignores the scope condition. The ablation refutes that: remove the Marker axis and the same model cites the scope condition and stays correct. The self-report described a preference, not a dependency. This is the chain-of-thought faithfulness problem [Turpin et al., 2023, Lanham et al., 2023, Chen et al., 2025] in miniature and with a cheap control: a one-line self-report, plus one ablation that removes the named field, separates what the model says it used from what it needs. We recommend the pairing (ask for the route, then delete the route and re-measure) as a routine check for any assay that reads a model's stated basis.

4.3 Relation to the Orientation Study

An earlier small study (four runs per cell, on a since-retired inference runtime with an unrecorded sampler) had found that adding structure to one decision class drove its firing scenario from three-of-four correct to zero, and read this as evidence of no reliable correct path beneath a shortcut. The present study, run clean and at larger scale, does not support that reading.

The earlier intervention *added* a routing block; ours *removes* the shortcut. When the shortcut is removed, the answer survives. We take the earlier collapse to have been an effect of the added structure, not evidence of a missing path, and we report the correction here because the orientation study's framing motivated this one.

4.4 Limitations

Eight runs per cell support statements about direction and about large gaps, not about small differences between counts. The field-source classifier keys on the first field-family term a response names; a minority of ablation-arm responses (31 of Qwen2.5-32B's 96) name a field the classifier records as "other" rather than resolving to a specific family, and we have not yet read each to its named field. Those are counted as non-Scope, which does not affect the direction of any result. The observable-and-evidence quality of each response (whether the cited condition is the discriminating one) is a separate, judge-scored layer we do not report here. The three subjects are small open-weight models on one GPU; we do not claim the exact rates transfer to larger or hosted models, only that the dissociation appears across two families, two scales, and both reasoning modes. Qwen2.5-32B ran at a smaller served context than the other two subjects; the prompts fit within it, so this bounds only unused context, but it is an inconsistency in the rig, recorded rather than hidden.

5 Conclusion

A structured decision rubric that names its decision field lets us separate three things a verdict-only assay treats as one: whether a model is right, which field it says made it right, and which fields it can use to be right. Across three small language models the first was near ceiling, the second was mostly off the intended field, and the third included the intended field throughout. Verdict accuracy did not track intended-field engagement, and the model's self-report described a preference that an ablation showed was not a dependency. A behavioral format assay should therefore measure the reasoning route, not only the verdict, and should check a model's stated route against an intervention rather than trust it. Attribution and capability are dissociable, and neither is read off the verdict.

5.1 Data Availability

The complete study (rubric-definition document, the six decision-class terrains in both the full and ablated forms, the twelve scenarios, the four-line instruction, the generated study definitions, every run's rendered prompt, response, and (for the thinking model) reasoning trace, the scored grids, and the scoring script) is available at `https://github.com/sophdn/corpos-lab` under the `studies/wrong-path-field-source/` directory.

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A Materials

The instruction, given to every subject after the assembled material, is reproduced verbatim:

> Read the glyph and the trace above.
>
> Respond using exactly this format — one line per field, no additional text:
>
> VERDICT: yes or no
> FIELD SOURCE: the name of the specific field in the glyph specification that determined your verdict
> OBSERVABLE: quote or closely paraphrase the specific condition from that field
> EVIDENCE: state which element of the trace your verdict rests on

The per-subject instruct wrappers, into which the assembled material is substituted, are: `[INST] {prompt} [/INST]` for Mistral; `<|im_start|>user\n{prompt}<|im_end|>\n<|im_start|>assistan` for both Qwen subjects, with no pre-filled think block so Qwen3.8 reasons.

B Decision Classes

The six decision classes are `cas` (companion-artifact-scope-gap), `cgu` (conditional-gate-uniform-default), `fsb` (formal-step-context-bypass), `gop` (governed-operation-protocol-bypass), `psc` (parent-state-check-bypass), and `scb` (structural-ceiling-bypass). Each terrain, in full and ablated form, and each scenario, is in the study directory named in Data Availability.